FCOM audit fixes

Small upstream bugs found by the FCOM audit of every A32NX and A380X system (2026-10-02): eleven A380X fixes, six A32NX fixes, the fast ADIRS alignment of both aircraft and the A32NX ND chronometer colour, each checked against the FCOM, an A32NX climb thrust regression test, and a fifth batch of FCOM gap fixes for both aircraft (ECAM lines, SD, brakes, GPWS, pressurization), the A380X FWS engine 4 take-off power fix, and the A380X fixes of 2026-10-12: door alerts and outflow valves for every door, L/G, NAV disagree and F/CTL law STATUS alerts, the exhaust heat haze of each engine and GPWS mode 4B, and one contrail per running A380X engine

Aircraft	A32NX and A380X
Where to find it	A380X: GPWS, OANS, SD (CRZ, PRESS, DOOR/OXY, F/CTL, G LOAD), FADEC, FWS (BRAKES HOT, T.O CONFIG, F/CTL ALTN LAW, HYD, CAB PRESS, DOOR, L/G, NAV), outflow valves, engine heat haze, contrails, ELEC panel, EWD EGT, brakes. A32NX: FWS (AIR PACK 2 OFF, ALL ENGINES FAILURE, F/CTL laws, ELAC FAULT, APU BLEED memo), BLEED, TCAS, SD (CRUISE, WHEEL, HYD), brakes, GPWS lights, ND chronometer. Both: ADIRS
Branch	`core/fcom-audit-fixes`, based on `master` (FlyByWire master 76db0aef1); independent of the other core branches. Each fix is small and can be taken upstream on its own. `core/failures-a32nx` merges this branch (its A32NX alerts build on the batch 5 F/CTL law lines and APU memos): merge it there again when this branch changes. Of the A380X FCOM batch 7, only GPWS mode 4B is here; its RCL / RCL LAST, T.O CONFIG NORMAL, automatic SD pages and BTV INOP parts are on `core/failures-a380x`. The twelfth A380X fix of the audit (SD FUEL jettison lines white during a normal jettison) needs the jettison of `core/fuel` and is in that branch. Two A32NX fixes of batch 5 (T.O CONFIG NORMAL with the wing tank pump alerts, FUEL L(R) WING TK LO LVL) need the fuel pump alerts and STATUS fields of `core/failures-a32nx` and are in that branch.
Status	Batches 1-4: all fixes tested in the sim by the user on develop (2026-10-04); not sim-tested on the master base, unit tests and Rust tests pass there. Batch 5 (2026-10-07): A32NX APU BLEED memo and SD HYD PTU colour, and A380X nose wheel steering with A-SKID OFF, tested in the sim by the user on develop: OK (2026-10-06); A380X SD F/CTL droop circles tested in the sim by the user on develop: OK (2026-10-08; G LOAD line hidden at normal g, its display outside 0.7-1.4 g not sim-tested); the other batch 5 fixes not sim-tested: unit tests and Rust tests. FWS engine 4 take-off power (committed 2026-10-08): a normal four-lever take-off on develop 2026-10-09 kept its phases (2, 3, 4, 6, 7, 8); the lever 3 / lever 4 split itself not sim-tested, unit test. 2026-10-12 fixes, tested in the sim by the user on the integration build before the commit: door alerts (D1, D2, N2), outflow valves with a door open (N5), heat haze (H1), L/G and NAV alerts (G1, V1, U1): OK (sessions of 2026-10-09/10); the F/CTL law STATUS lines and GPWS mode 4B not sim-tested (unit tests). Per-engine contrails: engine 1 flameout at FL350 stopped only its contrail, relit at FL290 282 kt (develop, test C1, 2026-10-12). None sim-tested on this core branch itself.
Pictures	No pictures: every fix changes a stock FBW display, alert or system value.

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1. A380X FCOM fixes

> **Aircraft:** A380X | **Where:** GPWS, OANS, SD, FADEC, FWS, ELEC panel, EWD | **Status:** Tested in the sim by the user on develop: OK (2026-10-04). Unit tests: LegacyGpws, OancSelectedRunway, FwsSystemDisplayLogicCrzPage, FadecWingAntiIce, BrakesHot,

> ElecPanelPbs, EgtLimits, CabinAltitudeIndication, FwsAbnormalSensedItems, OxygenIndication specs; Rust test in hydraulic/mod.rs (each fails without its fix).

What it is for

The FCOM audit of 2026-10-02 compared every A380X system with the A380 FCOM. These eleven findings were one or two wrong lines each (a variable nothing writes, a copy-paste index, an inverted condition). Each is now fixed as the FCOM describes.

What	Before	Now	FCOM
GPWS TOO LOW FLAPS	The FLAP MODE OFF selection was ignored (the GPWS read L:A32NX_GPWS_FLAP_OFF, which nothing writes)	TAWS FLAP MODE OFF on the MFD SURV CONTROLS page inhibits TOO LOW FLAPS	DSC-34-SURV
OANS selected runway	The FMS landing runway was compared with the FMS origin twice	The landing runway is green (selected) at the destination airport	DSC-34-10-70-20
SD automatic CRZ page	Engine 4 was not checked (engine 3 read twice)	Engine 4 alone at takeoff power also delays the CRZ page after takeoff	-
FADEC thrust limits	The wing anti-ice input read an A32NX variable that the A380X never writes	WING ANTI ICE ON (STRUCTURAL DEICE SWITCH) enters the thrust limit computation	DSC-70
BRAKES HOT and T.O CONFIG	Never triggered (L:A32NX_BRAKES_HOT is not written on the A380X)	From the 16 reported brake temperatures: one brake at or above 300 °C	PRO-ABN BRAKES HOT
ELEC panel COMMERCIAL 2 and ELMU pbs	All three pbs toggled COMMERCIAL (ELMU shed the commercial loads)	Each pb has its own state	DSC-24-20
EWD EGT amber limit	From the thrust lever angle	From the active thrust limit: none at FLEX/TOGA or in reverse, amber above the limit at MCT/CLB	DSC-70-90
SD PRESS CAB ALT needle	A320 delta P limits applied to feet	Red above 9 550 ft, like the value	DSC-21-30-20
F/CTL ALTN LAW procedure	MAX SPEED 310 KT shown only in ALT 1A (inverted)	Shown except in ALT 1A	PRO-ABN F/CTL ALTN LAW
HYD G RSVR LEVEL LO / G SYS OVHT	The G ELEC PMP A and B OFF line ticked from the yellow pumps; engine 3 and 4 pump disconnection on swapped DC buses	Green pumps; ENG 3 pump disconnection on DC 2, ENG 4 on DC 1	PRO-ABN HYD, DSC-29-40
SD DOOR/OXY crew supply	CREW SUPPLY OFF could not show; the CABIN line followed the crew oxygen pb	CKPT legend amber and REGUL PR LO on the ground with CREW SUPPLY OFF; the CABIN line amber on a low pressure only	DSC-35-20-20, DSC-35-30-20

Limitations and why

Limitation	Why	What to do
Only the wrong lines found by the audit were changed.	Each fix is kept to its finding so it can go upstream on its own; the rest of each system is as before.	-

ELMU and COMMERCIAL 2 have a state but no consumer yet.	Nothing in the electrical model sheds loads by those pbs (a380-simvars: A380X_OVHD_ELEC_COMMERCIAL_2_PB_IS_OFF, A380X_OVHD_ELEC_ELMU_PB_IS_OFF).	-
The EDP disconnection bus change has no cockpit test.	It only matters with DC 1 or DC 2 lost while disconnecting an engine pump; a Rust test covers it.	-

How to use

1. Nothing to do: the systems behave as in the FCOM.
2. TAWS FLAP MODE: MFD > SURV > CONTROLS, FLAP MODE OFF to inhibit TOO LOW FLAPS (e.g. a landing in a non-standard flap configuration).

How to test

#	Do this	Expected result
1	A380X on approach (FMS approach phase), gear down, flaps not in landing configuration, below 180 kt. MFD SURV CONTROLS: FLAP MODE on. Descend below 245 ft RA.	TOO LOW FLAPS.
2	Repeat with FLAP MODE OFF on the MFD SURV CONTROLS page.	No TOO LOW FLAPS.
3	Flight plan with a destination and a landing runway. At the destination, open the OANS map of the destination airport.	The FMS landing runway is shown green (selected runway).
4	After takeoff (climb), flaps and spoilers retracted, SD page not selected: keep thrust lever 4 alone at FLX/MCT or TOGA, levers 1-3 in CL.	The SD does not switch to the CRZ page at once: it waits for its 60 s timer, as with any engine at takeoff power. With all levers in CL it switches at once.
5	In flight, note the thrust limit on the E/WD; set WING ANTI ICE ON.	The thrust limit changes (the FADEC now sees the wing anti-ice).
6	After a high-energy stop (rejected takeoff or heavy braking), watch the SD WHEEL brake temperatures.	As soon as one brake reaches 300 °C: amber BRAKES HOT; the T.O CONFIG test then reports the hot brakes.
7	Overhead ELEC panel: press ELMU, then COMMERCIAL 2.	The COMMERCIAL pb stays as it was; each pb changes only itself.
8	Takeoff at FLX or TOGA, then thrust levers to CL.	EGT gauge: no amber EGT limit at FLX/TOGA; at CL the amber limit of the CLB thrust applies (EGT amber above it).
9	SD PRESS page with the cabin altitude above 9 550 ft (e.g. manual cabin altitude).	CAB ALT value and needle red.
10	Reach F/CTL ALTN LAW outside ALT 1A (e.g. with flyPad flight control failures).	The procedure shows MAX SPEED ..... 310 KT; in ALT 1A it does not.
11	flyPad > Failures > ATA 29: activate **Green reservoir leak**; when HYD G RSVR LEVEL LO shows, set the green electric pumps A and B OFF.	The procedure line of the green electric pumps A and B OFF ticks (the yellow pumps no longer tick it).
12	On the ground, overhead OXYGEN CREW SUPPLY pb OFF; SD DOOR/OXY page.	CKPT legend amber with REGUL PR LO; the CABIN line unchanged.

2. A32NX FCOM fixes

> **Aircraft:** A32NX | **Where:** E/WD, SD BLEED, CRUISE and WHEEL pages, TCAS | **Status:** Tested in the sim by the user on develop: OK (2026-10-04). Unit tests: DeltaPIndication, AutoBrakeIndication, AllEnginesFailure, TcasGpwsInhibit specs; Rust test in pneumatic.rs (each fails without its fix). The OPT RELIGHT SPD clean-up (2026-10-05): AllEnginesFailure spec; the E/WD line shown is unchanged.

What it is for

The same audit on the A32NX found six one-line bugs. Each now follows the A320 FCOM. The ALL ENGINES FAILURE procedure was also cleaned up on 2026-10-05, with the LEAP-1A relight envelope of core/failures-a32nx.

What	Before	Now	FCOM
AIR PACK 2 OFF	Checked the ENG 1 BLEED pb	Checks the ENG 2 BLEED pb (each pack against its own bleed)	PRO-ABN AIR PACK 1(2) OFF
BLEED LO TEMP flag	Set only with the bleed valve closed (inverted)	Set when the bleed supplies air (valve open) below 150 °C in flight with WING ANTI ICE on; the bleed temperature turns amber on the SD BLEED page	DSC-36
TCAS TA ONLY on a GPWS alert	Read L:A32NX_GPWS_Warning_Active (an A380X variable)	A GPWS warning or alert (L:A32NX_GPWS_WARNING_LIGHT_ON / ALERT_LIGHT_ON) inhibits the RAs (TA ONLY)	DSC-34-SURV-60-10-10
SD CRUISE delta P pulse	Pulsed during the whole cruise	Pulses only in flight phase 7 above 1.5 PSI, resets at 1 PSI	DSC-21-20-40
SD WHEEL AUTO BRK colour	Amber with an engine off (single-engine taxi, engine failure)	Green when armed; amber only on an AUTO BRK failure (not modelled)	DSC-32-30-20
ENG ALL ENGINES FAILURE: APU START line	Only below 2 500 ft radio altitude	Below FL 250 (pressure altitude), text unchanged	PRO-ABN ENG ALL ENGINES FAILURE
ENG ALL ENGINES FAILURE: OPT RELIGHT SPD	Three unused lines (280/.77, 300/.77, 260/.77) in the message table next to the one shown; the line list written inline in the FWC	One line, OPT RELIGHT SPD.270/.77 (the line already shown, the LEAP-1A value); the line list and its conditions in FWC/Logic/AllEnginesFailure.ts with specs	Air Arabia A320 QRH ABN-19.01A ALL ENG FAIL (LEAP-1A; CFM56: 300/.77)

Limitations and why

Limitation	Why	What to do
The APU START line does not say "(BELOW FL 250)".	The 24-character ECAM line has no room; the condition gates the line instead, as for the other conditional lines.	-
AUTO BRK never turns amber.	The AUTO BRK failure that the FCOM links to the amber colour is not modelled.	-

How to use

1. Nothing to do: the alerts and pages behave as in the FCOM.

How to test

#	Do this	Expected result
1	A32NX in cruise. ENG 1 BLEED OFF, ENG 2 BLEED ON, X BLEED closed. Set PACK 2 pb OFF.	After 5 s: AIR PACK 2 OFF.
2	In flight with WING ANTI ICE ON, at low engine power (cold bleed air); SD BLEED page.	When a bleed supplies air below 150 °C, its bleed temperature is amber; with that ENG BLEED pb OFF (valve closed) it is not.
3	In flight with a TCAS RA possible (traffic), trigger a GPWS alert (e.g. SINK RATE).	The TCAS goes TA ONLY (no RA) while the GPWS alert is on.
4	SD CRUISE page in cruise with delta P above 1.5 PSI.	Delta P does not pulse. In the approach (flight phase 7) above 1.5 PSI it pulses, until it is below 1 PSI.
5	On the ground, one engine running, AUTO BRK MAX armed; SD WHEEL page.	AUTO BRK green (it was amber).
6	In flight above 2 500 ft RA and below FL 250, APU off: all engines flame out (e.g. fuel cut-off).	ENG ALL ENGINES FAILURE shows the APU ..... START line, and one OPT RELIGHT SPD line: 270/.77.

3. A380X FCOM fixes, batch 5

> **Aircraft:** A380X | **Where:** SD F/CTL and G LOAD line, FWS (HYD, CAB PRESS), GPWS, pressurization, brakes | **Status:** Nose wheel steering with A-SKID OFF tested in the sim by the user on develop: OK (2026-10-06). SD F/CTL droop circles tested in the sim by the user on develop: OK (2026-10-08: no circles at flaps 0 and 1, circles at flaps 2, no G LOAD line at 0.9-1.1 g; the G LOAD line outside 0.7-1.4 g not sim-tested). The other lines: not sim-tested; unit tests AileronDroop, GLoadIndication, HydOverheatFlags, FwsAbnormalSensedItems and LegacyGpws specs, Rust tests in air_conditioning/mod.rs, hydraulic/mod.rs and autobrakes.rs.

What it is for

The fifth batch of the FCOM gap audit (reports of 2026-10-02) on the A380X: each line below differed from the A380 FCOM and now follows it.

What	Before	Now	FCOM
SD F/CTL aileron droop	The droop circle on the aileron scales in every configuration	Only in CONF 1+F, 2, 3 and FULL	DSC-27-10-20
SD G LOAD line	Shown at once whenever the value was outside 0.7-1.4 g, also on the ground; never XX	Only in flight, after more than 2 s outside 0.7-1.4 g; kept 5 s after the G load is back to normal; XX when the value is not valid	DSC-31-40-10
HYD G(Y) SYS OVHT	The two ENG PMP A+B DISC lines always shown	Shown only when turning the pumps OFF did not depressurise them	PRO-ABN HYD G (Y) SYS OVHT, DSC-29-20
HYD G(Y) SYS PRESS LO STATUS	BTV and CAT 3 DUAL missing; AVOID MAX TILLER ANGLE twice (Y); FUEL CONSUMPT INCRSD lines never shown	INOP SYS BTV and CAT 3 DUAL, INFO CAT 3 SINGLE ONLY; the Y STATUS shows TAXI WITH CARE and NO BRAKED PIVOT TURN; FUEL CONSUMPT INCRSD / FMS PRED UNRELIABLE when the PRIM also failed; Y SYS PRESS LO shows L/G GRVTY EXTN ONLY with the limitation MAX SPEED 220 KT (gravity extension)	PRO-ABN HYD G (Y) SYS PRESS LO

CAB PRESS EXCESS DIFF PRESS	Could trigger in flight phase 8 only	Can trigger on the ground before take-off and below 800 ft before landing (flight phases 1, 8 and 9)	PRO-ABN CAB PRESS EXCESS DIFF PRESS
GPWS mode 4 speeds	TOO LOW TERRAIN above 190 kt (4A) / 159 kt (4B), the A320 values, with a ceiling rising with the speed	Above 200 kt gear up (4A) and 180 kt flaps not in landing configuration (4B); a flat 500 ft (4A) / 245 ft (4B) ceiling while the terrain function is available, rising to 1 000 ft only with TERR SYS OFF or a TERR failure	DSC-34-20-20-10
Cabin descent rate	Limited to 350 ft/min	Limited to 300 ft/min	DSC-21-30-30
A-SKID switch OFF	Also cut the nose wheel steering (A320 logic)	The nose wheel steering stays: "A-SKID sw OFF does not cut off NWS"	DSC-32-30-10
AUTO BRK after take-off	Any mode disarmed 10 s after lift-off	Only RTO disarms 10 s after lift-off; a landing mode armed on the ground stays armed	DSC-32-10-30

Limitations and why

Limitation	Why	What to do
"Turning the pumps OFF was not successful" is read from the pump section pressure switch.	It is the only pump output the FWS has (design choice).	-
The GPWS "terrain function available" is the TERR SYS pb and the TERR failure of the selected system.	The FCOM names the aircraft position and the terrain database; the sim has no failure of either (design choice). The 1 000 ft ceiling is reached at 250 kt, as on the A320 (the FCOM figure is not precise enough to read the speed).	-
CAB PRESS EXCESS DIFF PRESS has no practical sim trigger.	It needs a differential pressure above the limit; the phase change is covered by the FwsAbnormalSensedItems spec.	-

How to use

1. Nothing to set: the pages, alerts and systems behave as in the FCOM.
2. A landing AUTO BRK mode can now be armed on the ground before take-off: it stays armed for the landing.

How to test

#	Do this	Expected result
1	A380X on the ground, aircraft powered, flaps lever at 0. ECAM control panel: **F/CTL** key. FLAPS lever to 1, wait; then to 2, wait.	Flaps 0 and 1: no white circles on the six ailerons. Flaps 2 (and 1+F on the way): a white circle on each aileron scale. The G LOAD line stays hidden (SD bottom, above the clock).
2	In flight, pull or push for more than 2 s outside 0.7-1.4 g (e.g. a firm pull-up above 1.4 g).	The G LOAD line shows the value; it stays 5 s after the load is back within 0.7-1.4 g. A shorter peak shows nothing.

3	Engines running, parking brake ON, ground speed 0. A-SKID switch (centre instrument panel, next to the L/G lever) OFF. Release the parking brake, taxi slowly and turn the tiller fully left.	The nose wheel turns and the aircraft turns left (before: no steering). Set A-SKID ON and the parking brake ON afterwards.
4	On the ground, AUTO BRK selector to a landing mode (e.g. LO). Take off.	The mode stays armed after lift-off (more than 10 s); with RTO armed, RTO disarms 10 s after lift-off.
5	In flight, flyPad > Failures > ATA 29: **Green reservoir leak**; wait for HYD G SYS PRESS LO. Open the STATUS.	INFO CAT 3 SINGLE ONLY; INOP SYS (APPR & LDG) includes BTV and CAT 3 DUAL.
6	Remove it; activate **Yellow reservoir leak**; wait for HYD Y SYS PRESS LO.	Procedure line L/G GRVTY EXTN ONLY; limitation MAX SPEED 220 KT (gravity extension) on the PFD and the ECAM; STATUS INFO TAXI WITH CARE, NO BRAKED PIVOT TURN, AVOID MAX TILLER ANGLE once, CAT 3 SINGLE ONLY.
7	A hydraulic system overheat (HYD G or Y SYS OVHT; set by the hydraulic model, this branch adds no failure for it): set the ENG PMP A and B pb-sws of the affected engines OFF.	The ENG PMP A+B DISC lines show only if a pump section keeps its pressure; with the pumps depressurised they do not show. Covered offline by HydOverheatFlags.spec.ts.
8	Approach over flat ground, gear up, flaps not in landing configuration, 220 kt, descend through 500 ft RA (MFD SURV CONTROLS: TERR SYS on).	TOO LOW TERRAIN from 500 ft RA, not higher. At 190 kt gear up: TOO LOW GEAR instead. (Gear up with the flaps in landing configuration: see "A380X GPWS mode 4B" below.)
9	Gear down, flaps not in landing configuration, 190 kt, descend through 245 ft RA.	TOO LOW TERRAIN from 245 ft RA (above 180 kt). Below 180 kt: TOO LOW FLAPS.
10	Repeat the gear-up pass at 250 kt with TERR SYS OFF.	TOO LOW TERRAIN from about 1 000 ft RA.
11	Descent at a high vertical speed; SD CAB PRESS page.	The cabin vertical speed goes down at no more than about 300 ft/min (was 350).

4. A32NX FCOM fixes, batch 5

> **Aircraft:** A32NX | **Where:** E/WD, SD HYD, brakes, GPWS/PULL UP pbs | **Status:** APU BLEED memo and SD HYD PTU colour tested in the sim by the user on develop: OK (2026-10-06). The other lines: not sim-tested; unit tests FlightControlLawAlerts, ApuMemos and PtuIndication specs, Rust tests in a320_systems (hydraulic brakes, lib.rs) and in the shared egpws.

What it is for

The fifth batch of the FCOM gap audit on the A32NX. The T.O CONFIG NORMAL fix (not given with a FUEL L(R) TK PUMP 1+2 LO PR alert) and the FUEL L(R) WING TK LO LVL fix (FUEL MODE SEL MAN only with fuel in the centre tank, and its STATUS) of the same batch are on `core/failures-a32nx`: they need its fuel pump alerts.

What	Before	Now	FCOM
F/CTL ALTN LAW	MAX SPEED and MANEUVER WITH CARE / MAX FL lines not as in the procedure; SPD BRK DO NOT USE not tied to an elevator fault	MAX SPEED 320 KT (320/.77 only with a dual hydraulic low pressure); SPD BRK DO NOT USE only with an L or R elevator fault; no MANEUVER WITH CARE, no MAX FL line	PRO-ABN F/CTL ALTN LAW
F/CTL DIRECT LAW	SPD BRK DO NOT USE next to USE SPD BRK WITH CARE	Only USE SPD BRK WITH CARE	PRO-ABN F/CTL DIRECT LAW

F/CTL ELAC 1(2) FAULT	FUEL CONSUMPT INCRSD and FMS PRED UNRELIABLE never shown	Shown after the procedure lines (hidden with them in the sidestick transducer case)	PRO-ABN F/CTL ELAC 1(2) FAULT
APU BLEED memo	Followed the APU bleed valve	Follows the APU BLEED pb-sw: APU available and pb-sw ON, at once	DSC-36-20
SD HYD PTU symbol	Amber also during the automatic PTU inhibitions (e.g. the second engine start with the parking brake set)	Amber only with the PTU pb-sw OFF	DSC-29-20
Alternate braking without anti-skid	Limited to 1 160 PSI	Limited to 1 000 PSI	DSC-32-30-10
Brake fans	Ran on the ground only	Run when the left main gear is down and locked, also in flight before landing	DSC-32-30-20
GPWS lights	The red PULL UP light for every GPWS alert	Red PULL UP only for the mode 1 PULL UP and mode 2 warnings; the other alerts (SINK RATE, DON'T SINK, TOO LOW ...) light the amber GPWS light of the GPWS/PULL UP pb	DSC-34-SURV-40

Limitations and why

Limitation	Why	What to do
An unused EGPWS test helper was removed in the same batch.	The Rust lint (clippy, all targets) failed on it; no behaviour change.	-

How to use

1. Nothing to set: the alerts, the SD and the systems behave as in the FCOM.

How to test

#	Do this	Expected result
1	A32NX on the ground, parking brake ON, APU running and AVAIL, APU BLEED pb OFF, engines off, PTU pb AUTO. Press APU BLEED ON, then OFF.	ON: the E/WD memo APU BLEED (green) replaces APU AVAIL at once. OFF: APU AVAIL comes back and APU BLEED goes at once.
2	APU BLEED ON. ECAM control panel: HYD page. Start engine 2 with the parking brake still ON (ENG MODE IGN START, ENG MASTER 2 ON), then press the PTU pb OFF and back to AUTO.	The PTU symbol stays green during the start (automatic inhibition); amber with the pb OFF, green again at AUTO.
3	In flight, flyPad > Failures > ATA 27: **ELAC 1**.	F/CTL ELAC 1 FAULT with its procedure lines and, at the end, FUEL CONSUMPT INCRSD and FMS PRED UNRELIABLE.
4	Remove it. ATA 29: **Green reservoir leak** and **Yellow reservoir leak**; wait for the G and Y pressures to be low.	F/CTL ALTN LAW (PROT LOST) with MAX SPEED 320/.77; no MANEUVER WITH CARE, no MAX FL 350, no SPD BRK DO NOT USE (no elevator fault).
5	When F/CTL DIRECT LAW shows.	USE SPD BRK WITH CARE, without a SPD BRK DO NOT USE line.
6	On the ground, A/SKID & N/W STRG switch OFF, press both brake pedals fully; triple brake indicator.	Brake pressure at most 1 000 PSI (was 1 160).

7	In flight on approach, BRAKE FAN pb ON, landing gear down and locked.	The brake fans run before touchdown (before: only on the ground).
8	In flight, trigger a GPWS alert other than PULL UP (e.g. SINK RATE, or TOO LOW GEAR).	The amber GPWS light of the GPWS/PULL UP pb on each pilot's panel; no red PULL UP. A mode 1 PULL UP or a mode 2 warning: red PULL UP.

5. ADIRS fast alignment in 30 s

> **Aircraft:** A32NX and A380X | **Where:** Overhead ADIRS panel | **Status:** Tested in the sim by the user on develop: OK (2026-10-04). Rust tests in adirs.rs (the existing fast alignment test tightened to 30 s).

What it is for

A fast (re)alignment, the IR mode selector set to OFF and back to NAV within 5 s, took 180 s. Both FCOMs give about 30 s (A320 FCOM DSC-22_20-20-40, IRS alignment page: "a fast alignment takes 30 s"; A380 FCOM DSC-34-10-30-30). The shared systems code now takes 30 s.

Limitations and why

Limitation	Why	What to do
Only the fast alignment changed.	A normal alignment from an unaligned IR is unchanged.	-

How to use

1. With the IRs aligned on the ground, set an IR mode selector to OFF and back to NAV within 5 s.

How to test

#	Do this	Expected result
1	Aircraft on the ground, IRs aligned (NAV). Set IR 1 to OFF and back to NAV within 5 s; start a stopwatch.	IR 1 is aligned again after about 30 s (not 3 min).
2	Repeat on the A380X.	Same: about 30 s.

6. A32NX ND chronometer in white

> **Aircraft:** A32NX | **Where:** ND, CHRONO pb on the EFIS control panel | **Status:** Tested in the sim by the user on develop: OK (2026-10-04). Spec ChronoColours.spec.ts (compiled styles of both aircraft).

What it is for

The ND chronometer was drawn in green digits on a light grey box, hard to read. The A320 FCOM gives "Chronometer Indication (white)" (DSC-31-45, INDICATIONS ON ND, item 9). The A32NX now shows white digits on a box in the display background colour.

Limitations and why

Limitation	Why	What to do
The dark box is a design choice.	The FCOM shows no box; a dark box keeps the digits readable over the map and terrain.	-
The A380X is unchanged.	Its FCOM gives no colour for the chronometer.	-

How to use

1. Press CHRONO on the EFIS control panel: the elapsed time appears at the bottom left of the ND. Press again to freeze it, a third time to reset and hide it.

How to test

#	Do this	Expected result
1	A32NX, ND powered: press CHRONO.	White digits on a black box at the bottom left of the ND, counting.
2	A380X: press CHRONO.	Unchanged: green digits on the grey box.

7. A32NX climb thrust regression test

> **Aircraft:** A32NX | **Where:** Unit tests (FMS and FADEC climb thrust tables) | **Status:** Test only, no code change: A320ClimbThrustLimit.spec.ts passes.

What it is for

A check of the A32NX climb against the A320neo FCOM climb tables (IndiGo FCOM PER-CLB-CLT, PDF pages 7248-7250 at ISA+10 and 7258-7260 at ISA+20, PW1127G-JM engines; no LEAP-1A table is available) found that the A32NX already climbs within 1-3 % of them, so no CLB thrust value was changed. The test keeps it so: the FADEC CLB N1 table (ThrustLimits_A32NX.hpp) and its FMS copy (cn1ClimbLimit) must stay identical row by row, and a climb computed with the FMS engine and flight models from FL 100 to FL 300 at 300 kt / M.78 must stay within 5 % of the FCOM times (8 % for each half) at 52 to 76 t.

Limitations and why

Limitation	Why	What to do
Nothing changes in the aircraft.	The finding was that the climb already matches the FCOM.	-
The reference is the PW1127G-JM table.	FBW models the LEAP-1A26; no LEAP climb table is available.	-

How to use

1. Nothing to do in the sim. When changing the CLB N1 table, change both copies and run the test.

How to test

#	Do this	Expected result
1	Run `npx vitest run fbw-a32nx/src/systems/fmgc/src/flightplanning/A320ClimbThrustLimit.spec.ts`.	All tests pass.
2	Optional in the sim: climbs at 52, 60 and 64 t, ISA+10, no wind, managed CLB speed, cruise FL 350; time from FL 100 to FL 300.	Within 5 % of 7.67 min (52 t), 9.08 min (60 t), 9.97 min (64 t); at ISA 2-5 % quicker.

8. A380X FWS flight phases: engine 4 take-off power from lever 4

> **Aircraft:** A380X | **Where:** FWS flight phases (take-off phase 3) | **Status:** No regression in the sim: a normal four-lever take-off on develop 2026-10-09 went through phases 2, 3, 4, 6, 7, 8 in order. The lever 3 / lever 4 split that shows the fix: not sim-tested. Unit test FwsTakeoffPower.spec.ts (fails against the previous FwsFlightPhases.ts).

What it is for

The FWS flight phases need two engines at take-off power for phase 3. The take-off power of engine 4 was read from thrust lever 3 (`eng4TLAFullPwr = eng3TLA > 43.3`), so lever 3 alone at TOGA counted as two engines, and lever 4 was never read. Each engine now reads its own lever; the rule (TOGA, above FLX/MCT, or FLX/MCT with a FLEX TEMP) sits in `FwsTakeoffPower.ts`.

Limitations and why

Limitation	Why	What to do
Nothing changes with all four levers moved together.	Only a lever 3 / lever 4 split shows the fix.	-

How to use

1. Nothing to set: set the thrust levers for take-off as usual.

How to test

#	Do this	Expected result
1	A380X on the ground, PARKING BRK ON, engines running at idle, all levers at IDLE, no FLEX TEMP. Thrust lever 3 only to TOGA, hold 5 s.	The FWS stays in phase 2: no red CONFIG PARK BRK ON (before the fix, phase 3 and that warning).
2	Lever 3 back to IDLE; levers 3 and 4 to TOGA.	Phase 3: red CONFIG PARK BRK ON.
3	All levers to IDLE.	The warning goes.

9. A380X door alerts and outflow valves for every door

> **Aircraft:** A380X | **Where:** E/WD, T.O CONFIG pb, SD DOOR and CAB PRESS pages, outflow valves | **Status:** Tested in the sim by the user on the integration build before the commit: DOOR UPPER 1L NOT CLOSED (test D1, 2026-10-09), DOOR FWD and AFT CARGO NOT CLOSED (D2, 2026-10-09), T.O CONFIG with an upper deck door open (N2) and the outflow valves held open (N5): OK (sessions of 2026-10-09/10). Unit tests: DoorAlerts.spec.ts; Rust tests in air_conditioning (cpiom_b.rs, mod.rs).

What it is for

The FWS and the pressurization read only some of the 18 A380X doors (10 main deck, 6 upper deck and 2 cargo doors). An open upper deck door or cargo door gave no alert, did not stop T.O CONFIG NORMAL and did not hold the outflow valves open. Every door now counts, as in the A380 FCOM (PRO-ABN-ECAM-10-52 DOOR alerts, PRO-ABN-ECAM-10-21 CAB PRESS INHIBITED BY DOORS, DSC-31-40-10 Takeoff Configuration Test, DSC-21-30-30).

What	Before	Now
DOOR ... NOT CLOSED cautions	Main deck doors only	Also **DOOR UPPER 1L** to **3R NOT CLOSED** and **DOOR FWD (AFT) CARGO NOT CLOSED**
IF ABNORM CAB V/S line of a DOOR alert	MAIN 5L/5R and UPPER 2L/2R showed the lines below it without the condition and never set the MAX FL 100/MEA limitation	A condition the crew confirms on every door; the MAX FL : 100/MEA-MORA limitation follows it
CAB PRESS INHIBITED BY DOORS	Not with MAIN 1R, the upper deck doors or the cargo doors	Any door not closed and locked at take-off power
T.O CONFIG test	Checked MAIN 1L, MAIN 2R and the cargo doors	Every main deck, upper deck and cargo door and the two cockpit sliding windows must be closed for T.O CONFIG NORMAL

Outflow valves with a door open at take-off power (AUTO)	The aft valves closed	The four outflow valves stay open (FCOM DSC-21-30-30: "maintaining the four outflow valves open")
Door inputs of the pressure controllers, cabin air and packs	Only MAIN 1L and 2R	All 16 passenger doors and both cargo doors

Limitations and why

Limitation	Why	What to do
No DOOR BULK CARGO NOT CLOSED.	The model has no bulk cargo door (no interactive point, no door in the systems).	-
A passenger door counts as not closed as soon as it moves.	The sim doors have no separate latch or lock.	-
The door alerts need an engine running.	They are inhibited in flight phase 1 (FCOM inhibitions).	-
A take-off with a door open keeps the outflow valves open until the next touchdown.	Design choice: the inhibition keeps its value from the lift-off. The manual pressurization mode is not affected (FCOM DSC-21-30-40).	-

How to use

1. Close every door and both cockpit sliding windows before the T.O CONFIG test.
2. With a DOOR ... NOT CLOSED caution, close that door (flyPad > Ground > Services, or the sim door controls) and follow the ECAM lines.

How to test

#	Do this	Expected result
1	A380X on the ground, PARKING BRK ON, engine 1 running at idle (engines 2-4 off), all doors closed. Open the upper deck door 1L (flyPad > Ground > Services: the second cabin door button at the bottom right; "Upper deck 1 L" on `core/efb-redesign`).	Amber **DOOR UPPER 1L NOT CLOSED**, MASTER CAUTION. Close the door: the alert goes.
2	All engines off, APU running. Open the forward cargo door (flyPad Services cargo door button), wait 30 s, then start engine 1.	With engine 1 running: amber **DOOR FWD CARGO NOT CLOSED** and **DOOR AFT CARGO NOT CLOSED**; SD DOOR page: FWD CARGO and AFT CARGO amber. (The cargo doors move only with all engines stopped: stop engine 1 to close them.)
3	All four engines running, flaps 1, FLEX entered (MFD PERF T.O), all doors closed. Open the upper deck door 1L; press the T.O CONFIG key on the ECAM control panel.	DOOR UPPER 1L NOT CLOSED; no T.O CONFIG NORMAL.
4	Door still open, PARKING BRK ON: ECAM control panel CAB PRESS key; all four thrust levers to FLX/MCT for 5 s, then IDLE.	SD CAB PRESS page: the four outflow valves stay fully open (before: the aft valves closed). Close the door afterwards.

10. A380X L/G, NAV disagree and F/CTL law STATUS alerts

> **Aircraft:** A380X | **Where:** E/WD, SD WHEEL and STATUS pages, PFD limitations | **Status:** Tested in the sim by the user on the integration build before the commit: L/G CTL faults on the ground (test G1) and BARO REF DISAGREE (V1): OK (sessions of 2026-10-09/10); L/G GEAR NOT LOCKED UP after take-off (U1): OK (2026-10-10). L/G GEAR NOT LOCKED DOWN, L/G DOORS NOT CLOSED, L/G SYSTEM DISAGREE, ALT / ATT / HDG DISAGREE and the F/CTL law STATUS lines: not sim-tested. Unit tests: FwsGearAlerts, FwsNavAlerts and FwsFctlAlerts specs.

What it is for

Round 8 of the FCOM gap audit: alerts of the A380 FCOM that the FWS did not have. The landing gear alerts come from the two landing gear control and indicating systems (LGCIS 1 and 2), the NAV alerts compare what the CAPT and F/O PFDs display.

Alert	When	What the crew sees
L/G CTL 1 (2) FAULT	One LGCIS failed	Crew awareness, SD WHEEL page, no chime
L/G CTL 1+2 FAULT	Both LGCIS failed, or invalid data from both	Single chime, MASTER CAUT; GPWS OFF line; NO L/G RETRACTION with the gear down, MAX SPEED 220 KT and gravity extension only with the gear up; STATUS INOP SYS (APPR & LDG) L/G CTL 1+2, GPWS 1+2
L/G GEAR NOT LOCKED DOWN	Lever DOWN, a gear not downlocked after about 60 s	Red warning with the recycle and gravity extension lines (gravity lines after 120 s); MAX SPEED 220 KT for the gravity extension
L/G GEAR NOT LOCKED UP	Lever UP, a gear not locked up after about 40 s	Single chime; recycle line (MAX SPEED 250 KT and AVOID EXCESS G LOAD with all gears down and a door open); STATUS INOP SYS L/G RETRACTION (L/G DOORS if a door is open), in flight INFO FUEL CONSUMPT INCRSD
L/G DOORS NOT CLOSED	A gear door not closed 30 s after the gear stopped moving (design choice: the FCOM gives no time)	MASTER CAUT, SD WHEEL page; STATUS INOP SYS L/G DOORS
L/G SYSTEM DISAGREE	The two LGCIS disagree for 5 s (design choice)	Crew awareness, SD WHEEL page
NAV CAPT AND F/O ALT DISAGREE	Displayed altitudes differ by more than 500 ft (STD) or 250 ft (QNH), same reference both sides	Single chime, MASTER CAUT; ALT ..... X CHECK, AIR DATA SWTG ..... AS RQRD
NAV CAPT AND F/O ATT DISAGREE	Pitch or roll differ by more than 5°	ATT ..... X CHECK, ATT HDG SWTG ..... AS RQRD
NAV CAPT AND F/O HDG DISAGREE	Headings differ by more than 5° (TRUE) or 7° (MAG)	HDG ..... X CHECK, ATT HDG SWTG ..... AS RQRD
NAV CAPT AND F/O BARO REF DISAGREE	One side in STD, the other on QNH	BARO REF ..... XCHECK
F/CTL ALTN LAW (PROT LOST), F/CTL DIRECT LAW	Existing alerts	New STATUS lines: INOP SYS (APPR & LDG) CAT 2 and GLS AUTOLAND; DIRECT LAW also LOAD ALLEVIATION, AP 1+2 and FD 1+2 (all phases) and INFO F/CTL BKUP CTL ACTIVE when all the PRIMs and SECs are failed

Limitations and why

Limitation	Why	What to do
The flight phase inhibitions are taken from the matching A320 alerts.	The FCOM gives them as figures only (design choice; the NAV alerts use the phases of the NAV ADR/IR FAULT alerts).	-
The NAV alerts wait 10 s.	The FCOM gives no confirm time; 10 s lets the two pilots set STD or a new QNH one after the other (design choice).	-
The model has three gears (nose, left, right).	The wing gears follow the body gears.	-
L/G GEAR NOT LOCKED UP needs a jammed gear.	The gear actuator failures (e.g. **Nose gear jammed actuator**) are on `core/failures-a380x`, not on this branch.	Merge it to test.

How to use

1. Nothing to set: follow the ECAM procedure when an alert shows, then read the STATUS page.

How to test

#	Do this	Expected result
1	A380X at the gate, engines running. flyPad > Failures > ATA 32: activate **LGCIU 1 Internal error**.	Amber L/G CTL 1 FAULT, no chime; SD WHEEL page.
2	Also activate **LGCIU 2 Power supply**.	L/G CTL 1+2 FAULT replaces it, single chime, lines GPWS OFF and NO L/G RETRACTION.
3	ECAM control panel: STS key.	INOP SYS APPR & LDG: L/G CTL 1+2, GPWS 1+2. Deactivate both failures.
4	On the ground, engines running. CAPT EFIS control panel: pull the BARO knob (STD); leave the F/O on QNH; wait 12 s.	NAV CAPT AND F/O BARO REF DISAGREE with BARO REF ..... XCHECK.
5	CAPT BARO knob back to QNH; wait 12 s.	The alert goes.
6	With `core/failures-a380x` merged: climbing above 1 500 ft, gear down, below 250 kt. flyPad > Failures > ATA 32: **Nose gear jammed actuator**; gear lever UP; wait 45 s.	About 40 s after the lever: amber L/G GEAR NOT LOCKED UP, single chime; STATUS L/G RETRACTION. Lever DOWN then UP: the recycle line ticks. Remove the failure, lever DOWN then UP.
7	In flight, reach F/CTL DIRECT LAW (flight control failures, flyPad > Failures > ATA 27); STATUS page.	INOP SYS: LOAD ALLEVIATION, AP 1+2, FD 1+2; APPR & LDG: CAT 2, GLS AUTOLAND.

11. A380X exhaust heat haze of each engine

> **Aircraft:** A380X | **Where:** Outside view behind the engines | **Status:** Tested in the sim by the user on the integration build before the commit: OK (test H1, 2026-10-09). Unit test: EngineEffectIndexes.spec.ts.

What it is for

The exhaust heat haze effects of aircraft.cfg read the wrong engines: the haze of engine 2 followed engine 1, and engines 3 and 4 followed engine 2 (an upstream bug). Each exhaust now reads its own engine.

How to use

1. Nothing to set: look at the engines from the outside view.

How to test

#	Do this	Expected result
1	A380X on the ground, engine 1 running, engines 2-4 off. Outside view of both wings from behind, 10 s.	Heat haze behind engine 1 (left outer) only; none at the right wing.
2	Start engine 2.	Heat haze behind engines 1 and 2 (left wing) only.

12. A380X GPWS mode 4B

> **Aircraft:** A380X | **Where:** GPWS aural alerts (mode 4) | **Status:** Not sim-tested. Unit tests: LegacyGpws.spec.ts.

What it is for

The A380 FCOM splits the GPWS mode 4 by the gear and flaps (DSC-34-20-20-10, a380_fcom.txt l.88328-88386): 4A with the gear up and the flaps not in landing configuration, 4B with the gear up and the flaps in landing configuration, or the gear down and the flaps not in landing configuration. Gear up with the landing flaps gave TOO LOW TERRAIN at high speed, and a mode 4 alert stayed on after the aircraft climbed back above its ceiling.

Gear / flaps	Now
Gear up, flaps not in landing configuration (4A)	Unchanged: TOO LOW GEAR below 500 ft under 200 kt, TOO LOW TERRAIN above 200 kt
Gear up, flaps in landing configuration (4B)	**TOO LOW GEAR** below 245 ft at any speed, never TOO LOW TERRAIN (the FCOM figure gives a flat 245 ft ceiling)
Gear down, flaps not in landing configuration (4B)	Unchanged: TOO LOW FLAPS below 245 ft under 180 kt, TOO LOW TERRAIN above 180 kt
Climbing back above the ceiling	The mode 4 alert stops

Limitations and why

Limitation	Why	What to do
Landing configuration = FLAPS FULL, or CONF 3 when CONF 3 is the landing configuration selected in the FMS.	As the existing FBW code (the flaps lever position, not the flap angle).	-
FLAP MODE OFF does not inhibit the gear-up 4B alert.	Design choice: the FLAP MODE pb is for a landing with an abnormal flaps configuration (FCOM l.88410-88412); only the flaps alert changes with it.	-

How to use

1. Nothing to set: the GPWS calls out as in the FCOM.

How to test

#	Do this	Expected result

1	A380X on approach (FMS approach phase), MFD SURV CONTROLS: GPWS and TERR SYS on. Flaps FULL, gear UP, 220 kt. Descend through 245 ft RA.	TOO LOW GEAR below 245 ft; no TOO LOW TERRAIN at any height.
2	Climb back above 245 ft RA (go-around).	The alert stops.

13. A380X contrails from each running engine

> **Aircraft:** A380X | **Where:** Outside view behind the engines, in cruise | **Status:** Tested in the sim by the user on develop: an engine 1 flameout at FL350 stopped only its contrail, the contrail came back after the relight at FL290, 282 kt (test C1, 2026-10-12). Unit tests: JettisonDripContrailEffects.spec.ts (7). Not sim-tested on this core branch itself.

What it is for

The fuselage part drew the contrails from the fuselage centre, whatever the engines did, and the wings part had a second copy on the exhaust nodes. Contrails come from the water vapour of the burnt fuel, so an engine that burns no fuel leaves none. The contrails are now declared once, on the exhaust of each of the four engines, and each shows only while the sim's contrail conditions are met and that engine burns fuel (MSFS engine combustion and a FADEC fuel flow above 0). A flamed out, shut down or seized engine leaves no contrail; the other engines keep theirs.

- One contrail per engine (engines 1 to 4), plus the ribbon contrail on engines 2 and 4 as before.
- Engine 2 now uses its own exhaust point (FX_EXHAUST_RIGHT2, the node of the left inner engine in the wing model despite its name); the old copy used the FX_EXHAUST_LEFT2 point.
- No contrail is declared in the fuselage part any more.

Limitations and why

Limitation	Why	What to do
Whether contrails form is decided by MSFS.	The effect uses the sim's CONTRAILS CONDITIONS MET (altitude and temperature), as the stock contrail templates do.	-
No N3 threshold for "running".	No source gives an A380 core speed limit for it; the fuel flow and the MSFS combustion flag are used instead (design choice).	-

How to use

1. Nothing to set: fly in contrail conditions (cold air at cruise level) and look at the engines from the outside view.

How to test

#	Do this	Expected result
1	A380X in cruise at FL350, all four engines running, contrails showing behind the engines. Outside view from behind.	One contrail behind each of the four engines, none from the fuselage centre.
2	Shut down engine 1: ENG 1 MASTER OFF.	The contrail of engine 1 (left outer) stops; engines 2, 3 and 4 keep theirs.
3	Descend to FL290 at about 280 kt and relight engine 1 (ENG START selector IGN/START, ENG 1 MASTER ON).	Once engine 1 runs again and contrails form at that level, its contrail comes back.